

# PROJECT REPORT

## Gaming\_Academic\_Performance

### 1. Introduction

Gaming Academic Performance is a data analytics project aimed at understanding behavioral and academic patterns among students. The dataset includes factors such as gaming habits, study time, sleep duration, stress levels, attendance, and grades.

The primary goal of this project is to analyze how lifestyle and gaming behavior affect academic performance.

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### 2. Problem Statement

Students today spend significant time using digital devices and playing online games. Excessive gaming can impact academic performance, mental health, sleep patterns, and productivity.

This project aims to:

- Understand gaming addiction behavior.
  - Analyze relationships between lifestyle factors and grades.
  - Discover trends using data visualization.
  - Provide meaningful insights for educational improvement.
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### 3. Dataset Description

#### Dataset Size

- Total Records: 8000
- Total Columns: 14

#### Data Types

- Numerical Features
- Categorical Features

#### Important Variables

- Gaming Hours
- Study Hours

- Sleep Hours
  - Attendance
  - Addiction Score
  - Stress Level
  - Grades
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## 4. Methodology

The project follows these steps:

### Step 1: Data Collection

Dataset collected in CSV format.

### Step 2: Data Preprocessing

- Checked for missing values
- Removed duplicates
- Verified data types
- Cleaned inconsistent values

### Step 3: Exploratory Data Analysis

Performed visual and statistical analysis using:

- Histograms
- Heatmaps
- Scatter plots
- Correlation analysis
- Box plots

### Step 4: Interpretation

Observed relationships among gaming habits, stress, attendance, and academic performance.

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## 5. Tools and Libraries Used

| Tool/Library | Purpose |
|--------------|---------|
|--------------|---------|

|                  |                           |
|------------------|---------------------------|
| Python           | Programming Language      |
| Pandas           | Data manipulation         |
| NumPy            | Numerical computations    |
| Matplotlib       | Data visualization        |
| Seaborn          | Statistical visualization |
| Jupyter Notebook | Development environment   |

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## 6. Data Analysis and Findings

### Dataset Summary

|                     |                |
|---------------------|----------------|
| Metric              | Value          |
| Total Records       | 8000           |
| Total Features      | 14             |
| Numerical Columns   | 11             |
| Categorical Columns | 3              |
| Missing Values      | Minimal / None |

The dataset mainly focuses on student gaming behavior and academic performance indicators.

#### 6.1 Gaming Hours Analysis

Students spending excessive time gaming generally showed lower academic performance and higher addiction scores.

#### 6.2 Study Hours Analysis

Students with higher study hours achieved better grades and maintained lower stress levels.

#### 6.3 Sleep Analysis

Adequate sleep duration positively affected stress management and academic productivity.

#### 6.4 Attendance Analysis

Attendance percentage strongly influenced academic success.

#### 6.5 Stress Level Analysis

Students with high stress levels tended to have:

- Lower sleep hours
- Higher device usage
- Reduced academic performance

## **6.6 Addiction Score Insights**

High addiction scores indicated:

- Increased gaming hours
  - Reduced study hours
  - Lower grades
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## **7. Visualization Summary**

The project generated multiple visualizations such as:

- Correlation heatmaps
- Gaming hours distribution
- Study hours comparison
- Grades distribution
- Stress level count plots
- Scatter plots between gaming and grades

These visualizations helped identify important behavioral patterns.

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## **8. Advantages of the Project**

- Helps understand student behavior.
  - Supports educational research.
  - Identifies unhealthy gaming habits.
  - Encourages balanced digital usage.
  - Useful for academic performance analysis.
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## **9. Limitations**

- Dataset may not represent all student populations.
  - Real-world behavior may vary.
  - Analysis is limited to available features.
  - No real-time tracking included.
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## **10. Future Scope**

Future enhancements may include:

- Machine learning prediction models
  - AI-based recommendation systems
  - Student performance forecasting
  - Real-time monitoring dashboard
  - Deep learning models for behavior prediction
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## **11. Conclusion**

The Gaming Academic Performance project successfully analyzed the relationship between gaming behavior and academic performance. The study revealed that excessive gaming and higher addiction scores negatively impact grades and stress levels, while study habits, sleep quality, and attendance positively influence academic success.

The project demonstrates the importance of data analytics in understanding behavioral trends and making informed educational decisions.

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## **12. References**

- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Kaggle Datasets
- GitHub Open Source Resources